

Course Outline

MAD 307 JavaScript Frameworks

Course ID: 030534**Academic Year:** 2025/26**Course Description:**

This course builds upon the fundamental knowledge learned in Javascript 1 through the introduction to a current front end web framework. Students will create real-world example web applications that can be deployed in a modern web environment. Following best-practice's, students will learn aspects of the specified framework and methods for organization and managing data in a scalable manner.

Pre-Requisites:	WEB 230	Category:	Vocational
Co-Requisites:	None	Course Credits:	3.00
Special Conditions:	None	Academic Level:	Credit (Post Sec)

Instructional Hours:	Classroom Instruction	30
	Laboratory/Workshops	15
	Other	0
Total Hours		45

Academic Department:

Windsor:	Zekelman School of Information Technology
Chair:	Ruth Susannah
Chatham:	
Chair:	

Revised By: M. Haug**Last Revision:** 2023/09/06

Required Tools, Equipment, and Learning Resources:

To be provided by instructor

Essential Employability Skills (EES):

Description	Teach	Assess
1) Communication: Communicate clearly, concisely and correctly in the written, spoken, and visual form that fulfills the purpose and meets the needs of the audience.		
2) Communication: Respond to written, spoken, or visual messages in a manner that ensures effective communication		
3) Numeracy: Execute mathematical operations accurately	✓	✓
4) Critical Thinking: Apply a systematic approach to solve problems	✓	✓
5) Critical Thinking: Use a variety of thinking skills to anticipate and solve problems	✓	✓
6) Information Management: Locate, select, organize, and document information using appropriate technology and information systems	✓	✓
7) Information Management: Analyze, evaluate and apply relevant information from a variety of sources	✓	✓
8) Interpersonal: Show respect for the diverse opinions, values, belief systems, and contributions of others		
9) Interpersonal: Interact with others in groups or teams in ways that contribute to effective working relationships and the achievement of goals		
10) Personal: Manage the use of time and other resources to complete projects		
11) Personal: Take responsibility for one's own actions, decision and consequences	✓	

Course Learning Outcomes (CLO):

Upon successful completion of this course, the student will be able to:
(EKS = Embedded Knowledge and Skills)

- Implement a framework deployment for creation of single-page applications to solve business problems. (CLO #1)
EKS:
 - Employ the use of framework tools, such as components, directives, pipes, routing, and services.
 - Implement data retrieval via asynchronous methods for single-page navigation.
 - Manage the navigation of the application with routing rules defined by the framework.

- Differentiate between the functionality in a single page versus multi-page application.
- Create a single-page application by employing an iterative workflow.
- Organize and manage data presented to an end-user through industry best practices. (CLO #2)
EKS:
 - Analyze the varying ways to accomplish the same outcomes in the framework.
 - Implement the best solutions for the right problems, to allow for the most efficient execution of the application.
 - Determine unexpected behaviours regarding certain aspects of the framework implementation, such as array mutation, data binding, and event handling, and how to avoid the problems they may incur.
- Implement efficient methods of rendering data using the framework to help optimize the end-user's experience. (CLO #3)
EKS:
 - Structure data in organized, logical data types, such as interfaces.
 - Utilize proper data binding to minimize how often the page needs to be re-rendered.
 - Implement the correct framework-specific tools to organize and display the correct data.
 - Outline why they are effective versus less complex methods such as using only primitives.
- Utilize framework tools to create a cohesive product, employing various structured languages to create robust web applications. (CLO #4)
EKS:
 - Identify the differences between the framework languages.
 - Implement the necessary steps to tie the different languages together into single, functional parts.
 - Apply different methods to organize parts of the framework to help organize the files logically.
- Construct applications that load data efficiently to minimize delays experienced by users, in contrast to traditional web applications. (CLO #5)
EKS:
 - Implement methods to retrieve data from a server using framework-built functionality.
 - Structure the display of data on the page in an efficient manner.
 - Implement methods to store some data locally to the user's machine.
 - Utilize more advanced methods of storing application data, such as local storage.

Teaching/Learning Activities:**Activities may include, but are not limited to:**

Discussion (large/small group, online disc. board)

Labs/Computer Labs

Lecture

Assessment:**Standard/Traditional Delivery**

- Assignments 40.00%
 - Frequency:* 10
 - Description:* Hands-on coding exercises which require students to complete small applications that focus on a specific topic area related to the framework including its syntax, constructs, and data structures. Students will follow conventions and best practices when applying their solutions. Each assessment is equally weighted.
 - Outcomes Assessed:* 1, 2, 3, 4, 5
 - EES Assessed:* 3, 4, 5, 6, 7
- Exam #1 25.00%
 - Frequency:* 1
 - Description:* Assessment covering material included in the first part of the course. It may include quiz questions and a practical programming portion where students will be required to program something new using the framework. Students will follow conventions and best practices when applying their solutions.
 - Outcomes Assessed:* 1, 2
 - EES Assessed:* 3, 4, 5, 6
- Exam #2 25.00%
 - Frequency:* 1
 - Description:* Assessment covering material included in the second part of the course. It may include quiz questions and a practical programming portion where students will be required to program something new using the framework. Students will follow conventions and best practices when applying their solutions.
 - Outcomes Assessed:* 3, 4, 5
 - EES Assessed:* 3, 4, 5, 6, 7
- Quizzes 10.00%
 - Frequency:* 10
 - Description:* Self-test online quizzes used to progressively evaluate the level of understanding of each topic covered in the course. Quizzes may include multiple-choice, multiple-answer, true/false, hot-spot, and short answer questions. Each quiz is equally weighted.
 - Outcomes Assessed:* 1, 2, 3, 4, 5
 - EES Assessed:* 3, 4, 5, 6, 7

100%

Note: The assessment listed in this outline represents the planned assessment method for this course. Unanticipated conditions during the delivery of the course may necessitate changes to the planned assessment. Students will receive reasonable advance notice should any changes be necessary.

Grading:

A = 80 - 100%
B = 70 - 79%
C = 60 - 69%
D = 50 - 59%
F = Less than 50%

Course Content:

See EKS statement

"Academic misconduct, including cheating of any form, will not be tolerated. Consequences may include, but are not limited to, a warning, a grade of "0" on the assignment/test/examination, or a failing grade in the course."

(Code of Students Rights and Responsibilities: Section 7.1.6)

All students and employees of this College have a right to study and work in an environment that is free from harassment and discrimination.

Accommodation Statement

The College will provide supports and services to all students with disabilities, both temporary and permanent, with valid supporting documentation. Interim accommodation requests will be received in good faith and can be provided pending receipt of medical documentation. Retroactive accommodations will be considered based on the unique circumstances of the individual matter. The College will give all Human Rights Code-related requests for accommodation meaningful consideration.

Procedure: The student is responsible to meet with a counsellor in Accessibility Services to discuss their functional limitations and accommodation needs and provide Accessibility Services with supporting documentation. Students are not required under the Ontario Human Rights Code to disclose their disability diagnosis (with the exception of Learning Disabilities) to receive accessibility supports and services and/or academic accommodations. Students are encouraged to meet with a counsellor prior to the start of a semester to provide information and arrange accommodations.